

Zakhar Pashkin

Senior AI Product Engineer | Computer Vision, OCR, Mobile/Cloud ML, Applied AI Delivery

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SUMMARY

Senior AI product and computer-vision engineer with 7+ years building OCR, segmentation, detection, mobile/cloud inference, and multimodal AI workflows. Owns the path from data quality and model evaluation to APIs, mobile/backend integration, QA evidence, and client-ready releases.

CORE STACK

Computer vision: OCR, segmentation, detection, landmarks, face texture analysis, multimodal retrieval, visual QA

Product engineering: FastAPI, REST APIs, TypeScript, React, Flutter/Dart, Firebase, Qdrant, Postgres, SQL

Modeling: Python, PyTorch, TensorFlow/Keras, OpenCV, YOLO, MediaPipe, CLIP/VLMs, ONNX Runtime, TFLite, CoreML

Delivery: Docker, Cloud Run, GCP, AWS, GitHub Actions, ClearML, MLflow, CI/CD, evals, release gates

EXPERIENCE

Senior Computer Vision & AI Product Engineer, Contract

Jun 2024 - Present

Dermaself & Gorilla GmbH - Contract, Remote

- Dermaself:** led CV product engineering for a skincare-analysis workflow across guided image capture, face-region processing, pore/wrinkle texture signals, result review, and mobile/backend integration.
- Defined API contracts, model-evaluation criteria, experiment tracking, QA checks, and mobile release boundaries for camera-heavy skincare analysis.
- Gorilla GmbH:** delivered CV/deep-learning work across dataset-quality review, model evaluation, inference behavior analysis, and implementation-ready technical recommendations.
- Managed contract delivery across requirements, milestones, acceptance criteria, client demos, QA evidence, and release notes.

Stack: Python, PyTorch, OpenCV, ONNX/TFLite, FastAPI, Flutter/Firebase, cloud deployment, Docker, GitHub Actions, MLOps, model evaluation, release QA

Senior Computer Vision Engineer

Jun 2022 - Jun 2024

Wombat Apps LLC / Carb Manager - Contract - Remote

- Built cloud and mobile ML workflows for food recognition, nutrition-label OCR, and structured nutrition extraction in Carb Manager product flows.
- Owned model-to-product integration across dataset preparation, model training/evaluation, backend APIs, mobile runtime constraints, and rollout support.
- Improved recognition reliability and mobile inference behavior through model export, input-pipeline cleanup, runtime profiling, and backend integration.
- Partnered with data, mobile, backend, and product teams on OCR/nutrition extraction quality, privacy constraints, and production reliability.

Stack: Python, Computer Vision, OCR, food recognition, mobile ML, cloud ML, API integration, model evaluation

Computer Vision Engineer

Jun 2019 - Jun 2022

CFT - Akademgorodok, Novosibirsk

- Built OCR, segmentation, detection, classification, and document-recognition models for fintech and mobile applications.
- Optimized energy-meter OCR and segmentation models for mobile inference through model architecture, quantization, and deployment-focused tuning.
- Delivered masked-face anti-spoofing and offline signature-verification prototypes with error analysis tied to deployment constraints.
- Created CLIP pre-validation and Toloka-assisted labeling workflows to reduce manual review load and improve dataset iteration quality.
- Mentored apprenticeship engineers inside the CFT program on CV project structure, dataset quality, error analysis, and deployment constraints.

Stack: Python, OpenCV, PyTorch/TensorFlow, OCR, segmentation, detection, mobile inference, data labeling

Machine Learning Specialist & Project Mentor

Oct 2018 - Oct 2019

School of AI - Part-time community ML program; ML Specialist (Oct 2018-Apr 2019), Project Mentor (Apr-Oct 2019)

- Built and maintained ML/DL learning software, including Flutter analytics/course surfaces and GCP-backed infrastructure for online education.
- Coordinated remote volunteers and contributors on platform work, course operations, and community program delivery.
- Turned TensorFlow and OpenCV topics into project-oriented learning paths and mentored participants on dataset setup, model iteration, and demo readiness.

SELECTED WORK

Dermaself Skincare Analysis

Mobile skincare-analysis work covering guided capture, face-region processing, pore/wrinkle texture signals, model evaluation, and release QA boundaries.

Gorilla GmbH CV Delivery

Client CV/deep-learning work covering dataset quality, inference behavior, model evaluation, and implementation-ready recommendations.

Multimodal Video Search Platform

Keyframes, ASR/OCR, object and face signals, CLIP-style embeddings, Qdrant/Postgres retrieval, and hybrid ranking checks.

Full-Face Wrinkle & Texture Lab

Cosmetic face-region masks, YOLO segmentation, skeletonized wrinkle traces, overlays, CSV outputs, timing events, and QA panels.

Fast OCR ONNX Inference Server

Containerized FastAPI OCR service with line segmentation, word segmentation, CRNN recognition, CPU ONNX runtime, and JSON text/box responses.

CV Repro Lab Tooling

Public reproducibility tooling for benchmark-gated CV experiments, browser validation, run cards, dataset manifests, and claim-review workflows.

EDUCATION & CREDENTIALS

- **Siberian Federal University** - Computer Science and Microelectronic Engineering, 2007-2012.
- **School of AI Dean's Credential** and Spring 2019 Fellowship Program training.
- **Professional coursework:** TensorFlow, deep learning and computer vision, SQL, JavaScript, Kubernetes on Google Cloud, and PyTorch scholarship work.